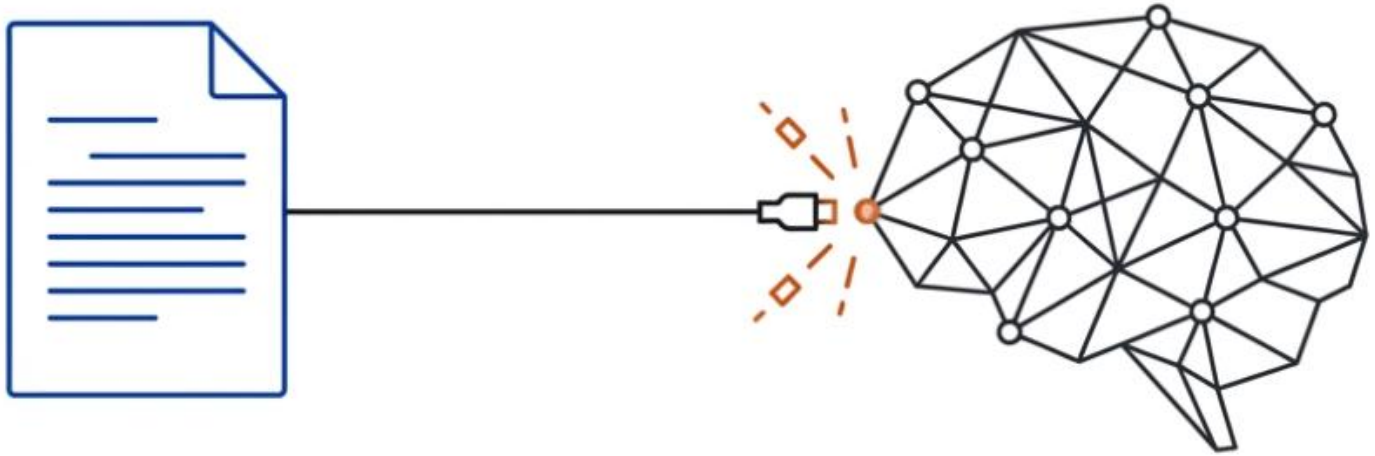
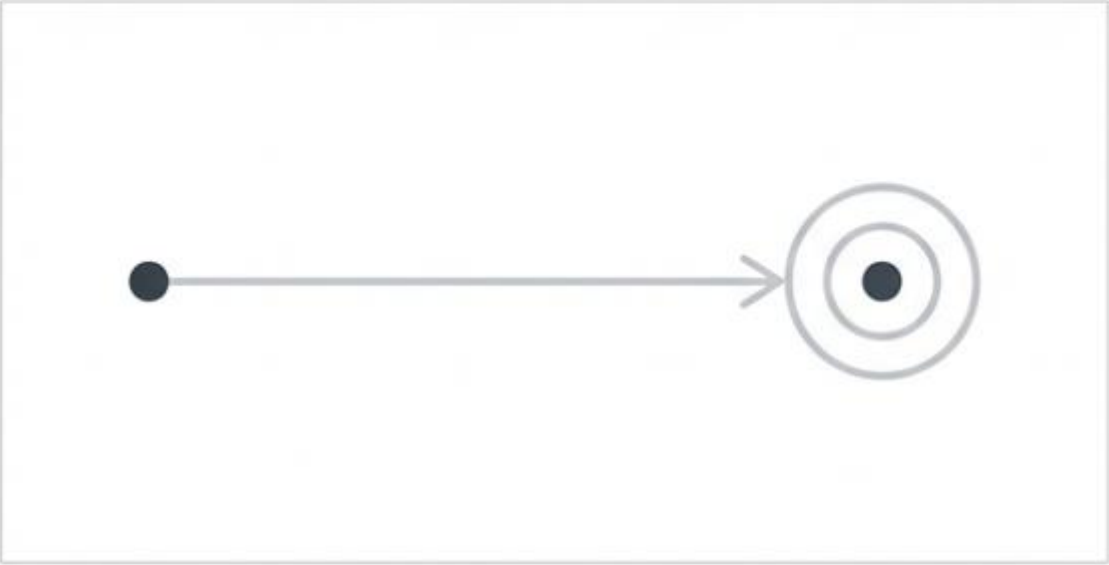


From Search to Answers: Building Your First RAG Pipeline

Chat history handling, memory strategies. Query Rewriting, Follow-up queries.

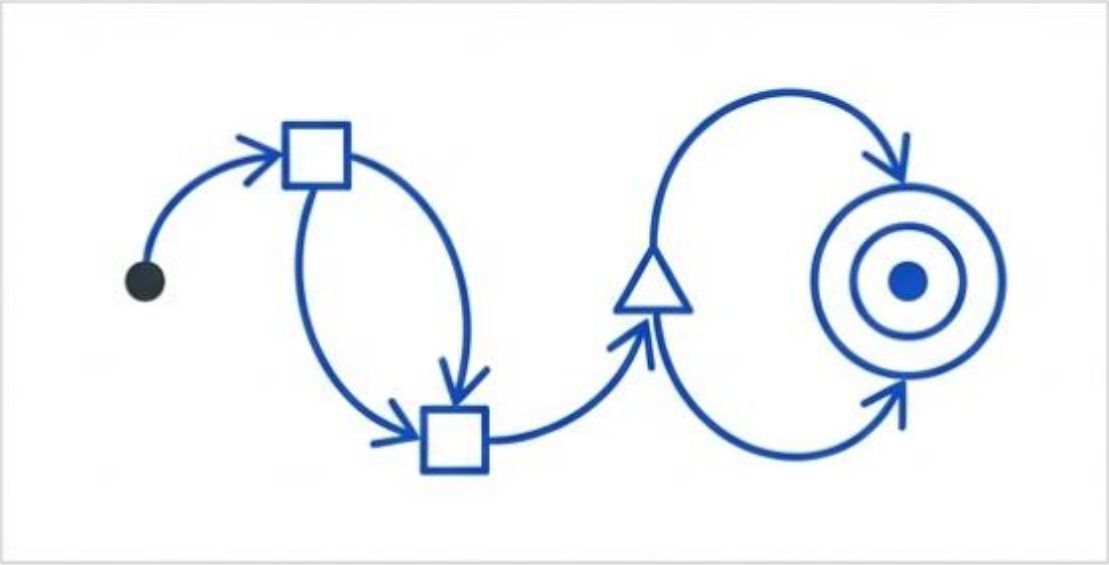
https://tinyurl.com/rag-augmented

The Core Shift in Architecture



RAG finds evidence for one query.

(Single-turn RAG can answer one question well, but assumes each question is entirely self-contained)



Conversational RAG keeps track of what the user really means across multiple turns.

(Sustains intent, resolves references, and maintains state)

Conversation history helps the assistant understand intent.
Query rewriting helps the retriever find evidence.

A Simple Conversation That Breaks Naive Retrieval

User: "How is EV charging infrastructure planned in India?"

Assistant: "[Answers comprehensively based on documents]"

User: "What are the main challenges?"

Assistant: "[Answers based on earlier context]"

User: "And what about the grid?"

Naive Retrieval Query:
"What about the grid?"

The last message alone is incomplete. Grid in what context?
Grid impact of EV adoption? Grid connection for charging stations?
A naive retriever will search literally for this fragment and return weak or irrelevant results.

The Anatomy of Memory

Memory is the stored conversation state used to maintain coherence across turns.
It is not about storing everything forever, but preserving what remains relevant.

Full Buffer

- **Pros:** Simple, lossless.
- **Cons:** Token heavy, older irrelevant turns pollute understanding.

Sliding Window

- **Pros:** Simple, efficient.
- **Cons:** Older important information is lost. Best for short chats.

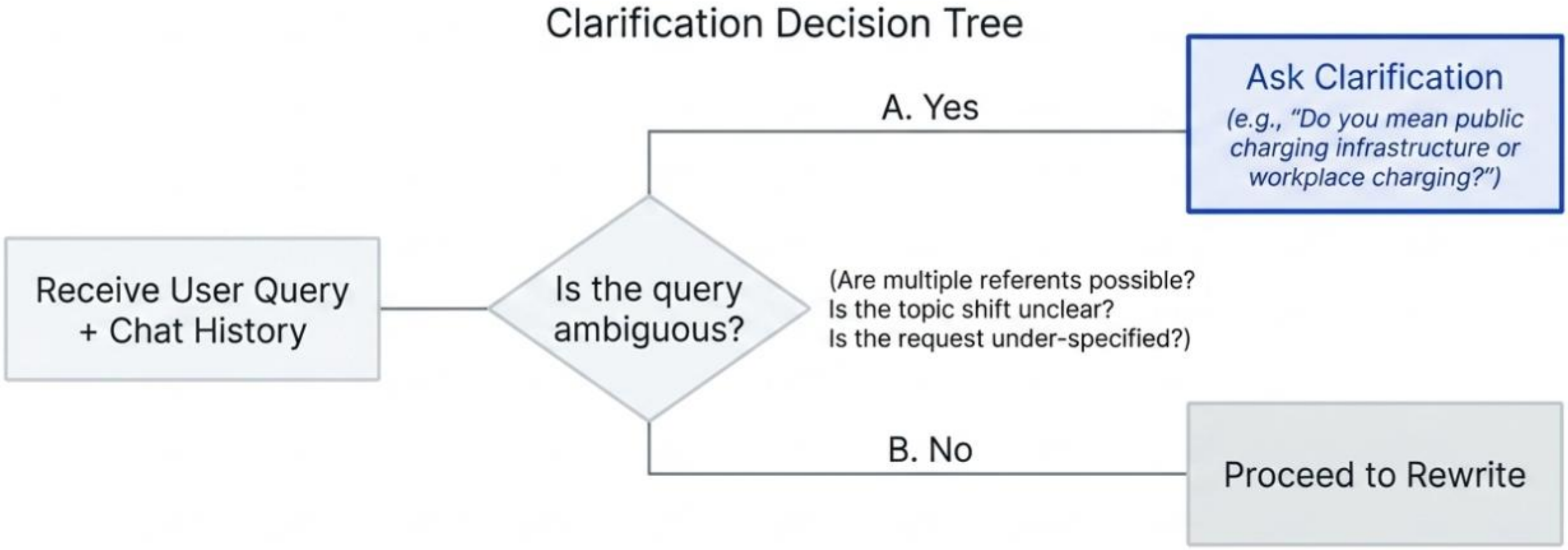
Summary Memory

- **Pros:** Keeps long-term gist, reduces token load.
- **Cons:** Loses detail, reliant on summarizer quality.

Hybrid Memory

- Keeps recent turns verbatim + summarizes older turns.
- Balances detail and efficiency. The most practical design.

Clarification is Better Than Guessing



Without clarification, the assistant may retrieve using the wrong intent and answer the wrong question fluently. Asking one good question prevents an entire incorrect answer.

Query Rewriting: The Bridge Between Chat and Search

User Follow-up:

How is that different from workplace charging?

Memory State:

(Topic: India EV documents; Active comparison target: public charging).

LLM Rewriter

Infers intent, recovers omitted context, resolves pronouns, restores the subject.

Rewritten Query:

How does workplace charging differ from public EV charging infrastructure in the India EV documents?

Lab

Thank You

Appendix